12. Wild plants — identifying, eating, using and avoiding poisoning

Oracle OS guide. Based on US Army FM 3-05.70 / ATP 3-50.21 (edible and poisonous plants, **illustrated** in the library), offline Wikipedia and Hesperian. Use **Identify a plant from a photo** (Library tab or Home → Plants) as a starting point, never as final confirmation.

Golden rules

1. **If in doubt, don't eat it.** Hunger kills in weeks; some plants kill in hours.
2. Confirm with **two sources** (photo + illustrated guide + someone who knows). Check **every part**: leaves (shape, arrangement, smell), stem (hollow? spotted? hairy?), flower, fruit, root.
3. **Mushrooms: never** from a photo or a "test". Several deadly ones (Amanita phalloides) look edible and symptoms come too late.
4. Avoid plants by roads, sprayed fields, polluted water and industrial land.
5. Children, pregnant women, the elderly and the sick are more sensitive.

Warning signs (avoid unless absolutely sure)

- Carrot family (**Apiaceae**: white umbrella-shaped flowers) — includes **hemlock and hemlock water dropwort**, the deadliest plants in Europe. Purple-blotched stem and mousy smell = hemlock.
- Milky sap (except dandelion and a few known ones), bitter-almond smell, leaves in threes (poison ivy in America), seeds in pods (many wild legumes are toxic), white or yellow berries, almond-like fruits.
- Strong bitterness or burning in the mouth → spit it out.

Classic dangerous look-alikes (Europe)

- **Wild garlic** (smells of garlic when crushed) vs **lily of the valley**, **autumn crocus** and **lords-and-ladies** (no garlic smell, deadly).
- **Wild carrot / parsnip** vs **hemlock**, **hemlock water dropwort** (Oenanthe — deadly, in ditches and riverbanks).
- **Comfrey** vs **foxglove** (digitalis, affects the heart).
- **Wild parsley/chervil** vs **fool's parsley** (Aethusa).
- **Elder** (ripe berries edible cooked; raw berries, leaves and stems toxic) vs **dwarf elder** (toxic).
- **Bilberries** vs **deadly nightshade** berries (shiny black with a 5-pointed "star" of sepals — deadly).

Common useful plants in the UK and Portugal (always confirm)

- **Nettle**: young leaves cooked (boiling removes the sting) — soup, tea; fibre for cordage.
- **Dandelion**: young leaves in salad (bitter), roasted root as "coffee", flowers.

- **Bramble/blackberry**: raw fruit; young leaves for tea.
- **Elder**: flowers for infusion/cordial; ripe berries **cooked only**.
- **Plantain**: young leaves cooked; crushed leaf on insect stings.
- **Clover**, **sorrel** (small amounts — oxalates), **purslane**, **wild mustards**.
- **Sweet chestnut** (boiled/roasted; don't confuse with **horse chestnut**, whose husk has short sparse spikes and nut has no point — toxic), **hazelnuts**, **acorns** (soak and boil in several changes of water to remove tannins), **stone-pine nuts**.
- **Rose hips**: rich in vitamin C, remove the hairy seeds.
- **Seaweed** from clean water: sea lettuce, laver/nori.

Universal edibility test (last resort only, never for mushrooms)

Takes about 24 hours per plant part. Only test abundant plants.

1. Fast for 8 hours (water only). Separate the plant into parts (leaf, stem, root, flower, fruit) and test **one part at a time**.
2. Smell: strong unpleasant, bitter-almond or peach smell → reject.
3. Rub on the wrist or inner elbow; wait 15 min. Burning, itching or redness → reject.
4. Prepare it as you would eat it (cooking is safer). Touch to the lip 3 min; then the tip of the tongue 15 min; then chew and hold in the mouth 15 min **without swallowing**. Any burning, numbness or bitterness → spit and reject.
5. Swallow a small amount and wait **8 hours** eating nothing else. Nausea, cramps → drink water, activated charcoal if you have it; inducing vomiting is generally not recommended.
6. No reaction: eat ¼ cup prepared the same way and wait another 8 hours. No reaction → that part, prepared that way, is probably safe.

Uses beyond food

- **Firewood**: birch (bark burns even damp), pine (resin = fire starter), oak and ash (long-lasting embers). Don't burn **yew**, **oleander** or **poison ivy** (toxic smoke).
- **Cordage/fibre**: nettle, inner bark of lime/willow, cattail (see "Ropes and knots").
- **Shelter**: conifer boughs for an insulating bed; dry bracken.
- **Traditional remedies** (variable effect, not a substitute for medicine): willow bark (salicylates, pain — avoid in children and with blood thinners), plantain (stings), chamomile and lime blossom (calming tea), thyme (cough). Doses and interactions: see "MSF Essential drugs" and "Where There Is No Doctor".
- **Insect repellent**: smoke from green leaves; crushed myrtle and eucalyptus.
- **Water**: tree leaves in a plastic bag in the sun (transpiration) give a little water; never from toxic plants.

Plant poisoning — what to do

- Remove remains from the mouth, keep a sample of the plant (for identification), call 999/112/NHS 111 or the poisons centre (Portugal **800 250 250**).
- Don't induce vomiting. Small sips of water. Recovery position if drowsy. CPR if breathing stops.

- Sap on skin (e.g. **giant hogweed** causes burns in sunlight): wash at once with soap and water, keep out of the sun for 48 hours.